COMPUTER ORGANIZATION AND ARCHITECTURE

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COURSE OBJECTIVES:	**3**	**0**	**0**	**3**

- To make students understand the basic structure and operation of digital computer
- To understand the hardware-software interface
- To familiarize the students with arithmetic and logic unit and implementation of fixed point and floating-point arithmetic operations
- To expose the students to the concept of pipelining
- To familiarize the students with hierarchical memory system including cache memories and virtual memory
- To expose the students with different ways of communicating with I/O devices and standard I/O interfaces

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PROCESSOR FUNDAMENTALS: 9

Computer Components - Performance Metrics - Instruction set architecture - Various addressing modes - Instruction execution in ALU - Simple data path

COMPUTER ARITHMETIC: 12

Representing unsigned and signed integer numbers - Floating point system - Integer addition and subtraction - Adders: Ripple carry adder, Carry Look Ahead adders - Integer multiplication and division - High-Radix Multipliers and High-Radix Dividers - Redundant number systems - Residue number systems

MEMORY SYSTEMS: 9

Memory hierarchy - Cache Memory: Organization, Design - Virtual Memory concepts

INTERCONNECTIONS AND PERIPHERALS: 6

Interconnection structures, Bus - PCI, Mesh, Hyper cube, Ring, Star - I/O Interface Systems: Keyboard, Monitor, Mouse, Bluetooth, USB, Flash

ILP ARCHITECTURES: **9**

Pipelining - Hazards in pipelining - Super pipelining - Super scalar - VLIW - Combining super scalar and VLIW with pipelining

TOTAL PERIODS: 45

COURSE OUTCOMES:

On successful completion of this course, the students will be able to

- Explain processor fundamentals
- Design arithmetic and logic unit
- Evaluate performance of memory systems
- Extend the learning to parallel processing architectures
- Explain interconnection structures

TEXT BOOK:

1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky, "Computer Organization", McGraw Hill Education, Fifth Edition, 2011.

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